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INTERIM FEDERAL SPECIFICATION

CLEANER, VACUUM, ELECTRIC

(INSTITUTIONAL/HOSPITAL)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The vacuum cleaners covered by this specification are of the electric motor powered vertical tank type to be used for wet and dry pickup in hospitals and institutions where aseptic conditions must be maintained. These cleaners are designed to operate on a 120 VAC, 60 Hertz, single phase power source.

1.1.1 Federal specification coverage. Federal Specifications do not include all varieties of the commodity as indicated by the title of the specification, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. The vacuum cleaners covered by this specification are of one type and shall be furnished in three sizes:

Size 1 - 5 gallon (liquid) capacity (minimum).

Size 2 - 10 gallon (liquid) capacity (minimum).

Size 3 - 15 gallon (liquid) capacity (minimum).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-600 - Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agengies).
Fed. Std. No. 141/6061 - Salt Spray (Fog) Test.

FSC 7910

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on s subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories (UL), Inc., Standard:

UL 1017 - Vacuum Cleaning Machines and Blower Cleaners.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, NY 11749; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

Floor and Vacuum Machinery Manufacturers Association (FVMMA), Inc., Standard:

Industry Standard No. 1 - Method of Rating Commercial and Industrial Type Vacuum Cleaners (Portable and Mobile) Suction Only.

(Application for copies should be addressed to the Floor and Vacuum Machinery Manufacturers Association, Inc., 1803 South Busse Road, Mt. Prospect, IL 60056.)

3. REQUIREMENTS

3.1 Materials. The materials employed in fabrication of these cleaners shall be sound, adequate for the intended purpose, and appropriately finished. All materials shall be free from defects which may affect the serviceability or the appearance of the vacuum cleaners.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the cleaners he proposes to supply under this specification conform to the requirements of UL 1017. The listing may be accepted as evidence that the cleaners conform to this requirement.

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3.2.2 In lieu of the listing, the bidder may submit proof, satisfactory to the contracting agency, that the cleaners conform to the applicable requirements of the published UL standards including methods of testing and test results.

3.2.3 Compliance with UL requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his cleaners. In event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

3.3 General. Each cleaner shall be complete, lubricated, and ready to operate (minor assembly is permitted). The design shall be simple in construction, shall permit easy accessibility to component parts for cleaning, inspection, adjustment, or replacement, and shall provide reliable operation. Similar component parts and assemblies of identical model number cleaners shall be fully interchangeable. Interchangeability shall not require alteration or modification.

3.3.1 Finish. All exterior surfaces, those interior surfaces impinged upon by the air flow and all surfaces subject to contact by the operator, shall be smooth and free from surface defects or obstructions which may contribute to dirt adherence or operator injury. Exterior surfaces shall be smooth, of uniform appearance, and suitably plated or finished to resist abrasion and corrosion. Interior surfaces shall be impervious to the corrosive attack of cleaning solutions at elevated temperature (180° F). Platings and finishes shall be in accordance with the best commercial practice.

3.3.2 Wet pickup. The cleaner shall be designed for wet pickup, i.e., retrieval of water or liquid cleaning solutions. The cleaner shall be provided with (a) automatic shutoff mechanism (see 3.6.4), (b) accessories required to convert the machine from wet pickup to dry pickup, and (c) dump valve (or similar mechanism) to drain the tank (see 3.6.3).

3.3.3 Duty. These cleaners shall be designed for continuous duty. Continuous duty is defined as constant operation for an 8-hour work day.

3.4 Identification marking. Each vacuum cleaner shall be plainly marked with the manufacturer's designation for the model or part number, and either his name or trademark of such known character that the source of manufacture may be readily determined.

3.4.1 Nameplate data. Nameplate data shall be prominently located. Data enumerated shall be in accordance with UL requirements.

3.5 Vacuum system.

3.5.1 General. The motor and the turbine or impeller shall be of rugged construction, shall operate at the minimum necessary r.p.m., shall be vibration isolated, and shall be attenuated for minimum noise. Air passages and housings shall be free from pits, flashings, sharp radii, etc., which may cause dirt impingement or build-up and turbulent flow.

3.5.2 Vacuum producer. The vacuum producer shall be of the centrifugal impeller type or multi-stage turbine type. The vacuum producer shall be balanced for minimum vibration. The vacuum producer shall develop the minimum water lift specified when tested in accordance with paragraph 4.3.3.5. All cleaners shall develop a minimum of 40 inches water lift through a 3/4 inch orifice and 8 inches water lift through a 1-1/2 inch orifice.

3.5.3 Vibration isolation. The motor and vacuum producer shall be mounted such that, while in operation, the assembly does not transmit vibrations to the body of the cleaner. Components of the cleaner shall not resonate when the cleaner is in operation.

3.5.4 Exhaust. The exhaust shall be peripherally expelled or expelled through openings of sufficient area such that exhaust velocity is at a minimum. The exhaust shall be expelled horizontally or at some angle to 60° above the horizontal to minimize possible exhaust disturbance of uncleaned areas. The exhaust system shall be suitably muffled to minimize noise (see 3.5.5).

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3.5.5 Noise. The maximum noise generated by the cleaner when in operation shall be 60 dbA. Measurements shall be taken with the test setup and procedure of 4.3.3.3.

3.5.6 Switch. The switch shall be a reliable, heavy duty quick make and break type. The switch rating shall be no less than the current rating on the motor nameplate and shall be UL listed. The switch shall be prominently and conveniently located.

3.5.7 Cord. The power cord assembly shall be 35 feet (minimum), three conductor, in accordance with UL. At the point of cord entry into the machine, a heavy duty strain relief shall be provided.

3.5.8 Cord storage. Each cleaner shall provide some means for coiling, wrapping, or otherwise storing the power cord when the cleaner is not in use.

3.6 Filtration system.

3.6.1 General. The filtration system shall consist of the filter assemblies and the tank assembly. When the cleaner is employed for dry pickup, all filter assemblies shall be employed. When the cleaner is converted for liquid retrieval, the separator and automatic shutoff assembly may take the place of the primary filter(s).

3.6.2 Filters. The primary filter shall filter gross particulate approximately 1 micron and larger and shall retain this matter for easy disposal. This filter may be disposable. The additional filter(s) shall be an impaction type and shall entrap all matter and micro-organisms 0.3 micron and larger. No ingested unfiltered air shall be exhausted from the cleaner.

3.6.3 Tank. The tank shall be of stainless steel when specified (see 6.2). Tanks, other than stainless, shall be suitably finished or plated to resist corrosion. Tanks of non-metal construction may be employed if service life and durability can be demonstrated to be equal to the metal tank (see 4.3.2.3). If weldments are used in tank construction, electrode or filler rod material shall approximate the base metal composition. Platings, coatings, or finishes used for corrosion protection shall resist the corrosive effects of normal cleaning solutions at 180° F. The tank shall have a convenient and efficient dump valve or similar mechanism to permit draining of liquids without requiring lifting of the tank.

3.6.4 Wet pickup. When the cleaner is to be used for liquid retrieval, the only filter which must remain installed is the 0.3 micron impaction filter. The machine shall be equipped with an effective separator which will prevent the impeller from ingesting liquids or solids retrieved. An automatic shutoff assembly shall be provided which interrupts the air intake or power to the cleaner when a pre-determined level is reached. A foam suppressor or a foam expansion space shall be provided on the tank. All components shall be fabricated from materials impervious to corrosive effects of cleaning solutions at elevated temperatures (180° F).

3.7 Mobility. The cleaner shall have a bearing equipped wheeled chassis or bearing equipped wheels mounted on the tank to provide mobility. Wheels and/or casters are acceptable in any combination. Wheel/caster configuration shall be such to permit the cleaner, when empty, to be tilted 10° off vertical (minimum) without toppling. The cleaner shall also be equipped with a handle assembly or hand holds to facilitate cleaner movement.

3.8 Attachments.

3.8.1 General. The cleaner shall be furnished with one dusting brush, one wall brush, one 14 inch (minimum) squeegee tool, one 12 inch (minimum) bare floor tool, one crevice tool, one (or sectional) wand assembly, and one 10 foot (minimum) hose assembly. The automatic shutoff assembly and the necessary attachments for conversion from dry to wet pickup shall be furnished as accessories with each cleaner (see 3.3.2). Variations may be specified (see 6.2).

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3.8.2 Tool construction. The tools shall be fabricated from synthetics or non-ferrous metals. The brush and squeegee inserts shall be replaceable. The tools shall mechanically lock on the wand assembly or the hose assembly (if a metal connector is employed). A friction fit between the tool and the hose is acceptable if the hose end is of rubber or synthetic.

3.8.3 Hose construction. The hose shall be fabricated from non-marking materials impervious to attack from common cleaning solutions. The hose shall be a minimum 1-1/2 inch inside diameter by 10 feet in length. A swivel coupling shall be provided on the hose end attaching to the cleaner. The use of a swivel coupling on the wand end of the hose is optional. Strength requirements of the hose and the couplings are defined by the tests of 4.3.2.5 and 4.3.2.6.

3.9 Manuals. Each cleaner shall be shipped with a manual or handbook of operating and maintenance instructions. The operating instructions shall explain the function of the major components with schematics, where necessary, and the maintenance instructions shall contain an Illustrated Parts Breakdown with replacement part numbers identified.

3.10 Workmanship. The cleaner and all tools and attachments shall be constructed, assembled, and finished in a manner to assure good quality equipment of a neat general appearance and free from imperfections which may affect durability or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2.2 End item inspection.

4.2.2.1 Visual examination. Examination of the end item shall be in accordance with the classification of defects set forth in table I. The lot sizes shall be expressed in units of vacuum cleaners for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely fabricated vacuum cleaner of each size offered for delivery at one time. The sample shall be visually examined for general defects identified in table I. The inspection level shall be level II of MIL-STD-105 using an Acceptable Quality Level (AQL) of 4.0 defects per one hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Finish not as specified:		
	Affects appearance.		X
	Affects serviceability.	X	
Design	Characteristics of machine not in accordance with the specified requirements.	X	
Construction and Workmanship	Components malformed:		
	Affects appearance.		X
	Affects serviceability.	X	
	Components improperly assembled:		
	Affects appearance.		X
	Affects serviceability.	X	
	Wiring cut, abraded, improperly staked, or connections loose.	X	
	Wiring strain relief inadequate, serve through holes not grommeted.	X	
General	Improper marking.	X	
	Inadequate manual.	X	

4.2.2.2 Dimensional examination. Inspection shall be made of the end item to determine compliance with dimensions specified. Any deviation from specified dimensions shall be classified as a defect.

4.3 Testing.

4.3.1 General. The tests specified may be performed in any order convenient to the contractor. One cleaner shall be selected at random for the following tests. Rejection of any cleaner in the lot for failure to comply with a specific test requirement shall result in rejection of the lot until the failed cleaner has been repaired and passes all tests. In addition, the test which resulted in failure of the sample shall be successfully completed by a second sample selected by the Quality Control Representative.

4.3.2 Mechanical preoperational tests.

4.3.2.1 Accelerated corrosion tests. The coatings applied to the interior of metal tanks, other than stainless steel, shall be subjected to a 1,000 hour corrosion test described in Fed. Std. No. 141, Test Method 6061 to determine suitability of the coating. The coating subjected to this test shall be impacted after the corrosion test. The coating shall exhibit good bonding. Any separation or fracture shall be cause for rejection.

4.3.2.2 Motor tests. The motors shall be tested in accordance with the requirements of UL 1017. The supplier of the cleaner may offer a test certificate from a testing laboratory or a statement of compliance from the motor manufacturer that the motors conform to and have been tested and satisfactorily meet the requirements.

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4.3.2.3 Non-metallic tank tests. Non-metallic tanks shall be tested at two temperature extremes of 40° and 180° F.

4.3.2.3.1 40° test. The tank shall be chilled to 40 ±5° F. after being filled with 5 pounds of sand or similar granular substance. The tank shall be raised approximately 2 feet above a concrete or ceramic floor and dropped on the edge of the tank bottom. If the tank is of rectangular form, the drop shall be made on the corner of the tank. Any indication of cracking or any fracture in the tank shall be cause for rejection.

4.3.2.3.2 180° test. The tank shall be filled to its rated capacity with 180° F. tap water. It shall be demonstrated that when raised by the hand grips to the height of a deep sink rim (30 ±6 inches) and rotated to a dump position, the tank maintains its configuration. Rim or grip deformation during this demonstration, even if the tank returns to its original shape, shall be cause for rejection.

4.3.2.4 Mobility tests. The cleaner shall be demonstrated to be free wheeling and easily transportable. Handles or grips shall be subjectively evaluated for comfort and location. The fully assembled cleaner shall be demonstrated to be stable. With the recovery tank empty, the cleaner shall be tilted to 10° off vertical in those directions perpendicular to a line drawn between the wheels or casters (see figure 1).

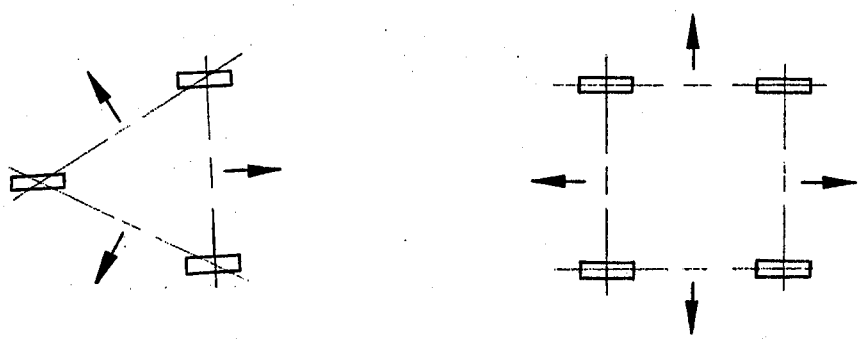


FIGURE 1.

4.3.2.5 Hose test. A compressive load of not less than 200 pounds shall be applied to any portion of the hose (no closer than 12 inches from any fitting) by means of parallel surfaces having a length of not more than 3 inches parallel to the axis of the hose, for not less than 2 minutes, and then the load released. There shall be no rupture of any portion of the hose and any resulting diameter shall not differ more than 1/16 inch from the diameter prior to the test.

4.3.2.6 Coupling test. A tensile force of not less than 25 pounds shall be applied to the engaged cleaner hose and tools for not less than 2 minutes and then the force released. Couplings shall engage and release satisfactorily.

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4.3.3 Operational tests.

4.3.3.1 General. The operational tests shall be performed with the cleaner connected to a nominal 120 volt, 60 Hertz, 1 phase power source. Each sample cleaner shall be operated in the condition received for 10 continuous hours. The orifice shall be wide open for this run. All lubrication points shall be visually checked after this run. Lubricant quality and quantity shall be visually analyzed. The lubricant shall not contain any metallic particles; the lubricant quantity shall be ample. Evidence of galling or similar deficiency or lack of lubrication shall be cause for rejection. The commutator(s) shall be visually examined. There shall be no evidence of burning or pitting.

4.3.3.2 Exhaust diffusion tests. Place the cleaner in an operating position with the intake hose detached. Encircle the cleaner with a suitable wire mesh screen to which are attached cotton or wool yarn streamers. These streamers shall be attached in random fashion in an area from the horizontal plane of the cleaner exhaust ports to 12 inches below this plane. The horizontal distance from the cleaner exhaust to the screen shall be approximately 2 feet. The streamer free length shall be approximately 2 inches. With the machine in operation, it shall be observed that no streamer is disturbed by the exhaust air flow.

4.3.3.3 Noise tests. The cleaner shall be set up in operational configuration in an anechoic chamber. The wand assembly and hose shall be coupled to the cleaner and essed so the wand is approximately 45° to the vacuum inlet. The microphones shall be located approximately as shown. Microphone 1 shall be 5 feet above the floor and microphone 2, 3 feet above the floor. Laboratory quality calibrated sound recording equipment shall be used for pickup and recording of the cleaner noise generated. Slow response or damped indicator readings are the only readings to be recorded. The maximum noise levels generated shall be 60 dbA (see figure 2).

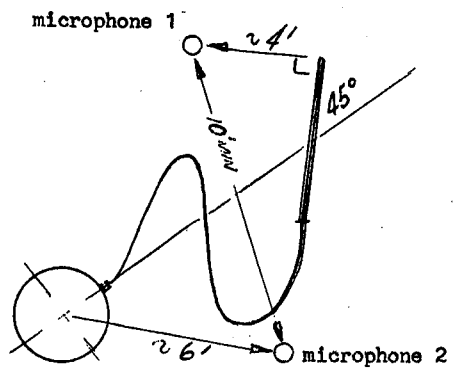


FIGURE 2.

4.3.3.4 Filtration tests. The contractor shall certify, or shall contract an independent laboratory to certify, that the filter system entraps and retains all micro-organisms 0.3 micron and larger ingested by the cleaner. This certification shall detail the test methods, the aerosols, the culture mediums, the sample count, the bacterial count of the ingested air, the bacterial count of the exhausted air, and the collection efficiency expressed as a percentage. Culture mediums shall be impinged upon by the exhaust air for 5 ±.5 minutes for the efficiency samplings. Certification tests shall also demonstrate that organisms cannot migrate through the filters. This shall be demonstrated by sampling exhaust air for at least 30 minutes after shut down of the bacterial aerosol which the cleaner had been ingesting for a prior 30 minutes. Clean, new filter efficiencies shall be a minimum of 99.5 percent. Collection efficiencies shall show an upward trend as the cleaner is used.

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